

A decorative graphic on the left side of the slide consists of two overlapping parallelograms. The front one is a vibrant blue, and the one behind it is a light green. They are positioned diagonally, creating a sense of depth and movement.

SCTP DSAI Module 2 Project Presentation

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Business Case

Leverage historical transaction data to identify popular products, match consumers with relevant sellable listings, and increase product visibility to drive sustained platform engagement and profitability

Two-Layer Data Platform

Separate ingestion from analytics, then scale storage and compute on demand.

THE STACK AT A GLANCE

LAYER 1 • INGESTION

Meltano

Open-source ELT – no licences, no lock-in

9 SOURCE FILES – 7 IN SCOPE

Olist e-commerce CSV
extracts loaded by Meltano

- ✓ olist_customers_dataset.csv
- ✓ olist_orders_dataset.csv
- ✓ olist_order_items_dataset.csv
- olist_order_payments_dataset.csv
- ✓ olist_order_reviews_dataset.csv
- ✓ olist_products_dataset.csv
- product_category_name_translation.csv
- olist_geolocation_dataset.csv
- olist_sellers_dataset.csv

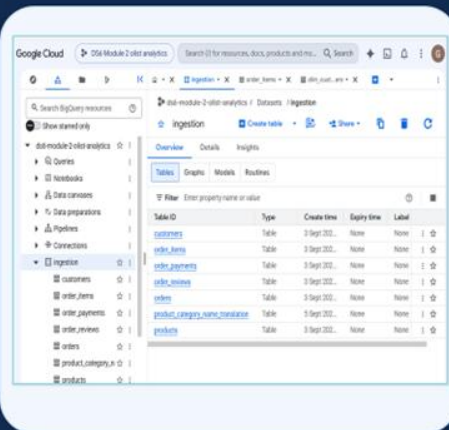


LAYER 2 • STORE +

ANALYZE

GCP BigQuery

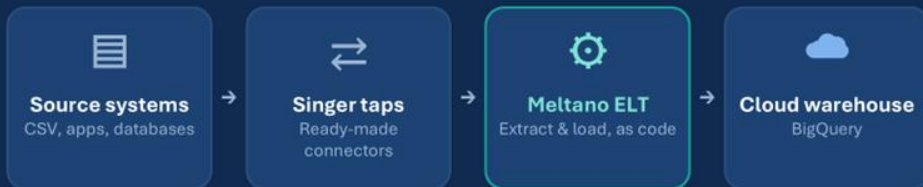
Self-managing warehouse – elastic scale



Sources → Meltano → BigQuery → Dashboards & insights

Ingestion Layer – Meltano

Open-source ELT – pipelines as code, no licences, no lock-in.



WHY MELTANO



INGESTION
Meltano

Open-source ELT / DataOps

- ✓ \$0 · no licence cost
- ✓  no vendor lock-in
- ✓  new source in days
- ✓  dev / staging / prod as code

Data Warehouse – GCP BigQuery

Self-managing cloud warehouse – elastic scale, standard SQL.



Storage

Pay for what we keep



Compute

Pay for what we query



GCP BigQuery – one trusted analytics layer

\$0.00 / month today – inside the free tier



WHY BIGQUERY



STORE + ANALYZE

GCP BigQuery

Self-managing cloud warehouse

- ✓ ⚙️ zero maintenance – no servers
- ✓ ⬆️⬆️ storage & compute scale apart
- ✓ SQL · productive from day one
- ✓ 🛡️ Google Cloud governance
- ✓ 💰 pay only for what we query

Advantage: enterprise-scale analytics at near-zero run cost

BigQuery Running Cost — \$0.00 / Month

Two independent cost components, both inside Google's permanent free tier.

Olist CSV files

~200 MB, 9 tables



Meltano ELT

load to warehouse



**BigQuery
storage**

0.2 GB at rest



Queries & dbt

a few GB scanned / mo



Dashboards

insights

STORAGE — DATA AT REST

\$0.00

/ month

Our data: 0.2 GB

Free tier: first 10 GB each month

List price beyond free tier: \$0.02–\$0.023 per GB



2% of the free allowance used

COMPUTE — DATA SCANNED

\$0.00

/ month

Our usage: a few GB scanned per month

Free tier: first 1 TB scanned each month

List price beyond free tier: \$6.25 per TB

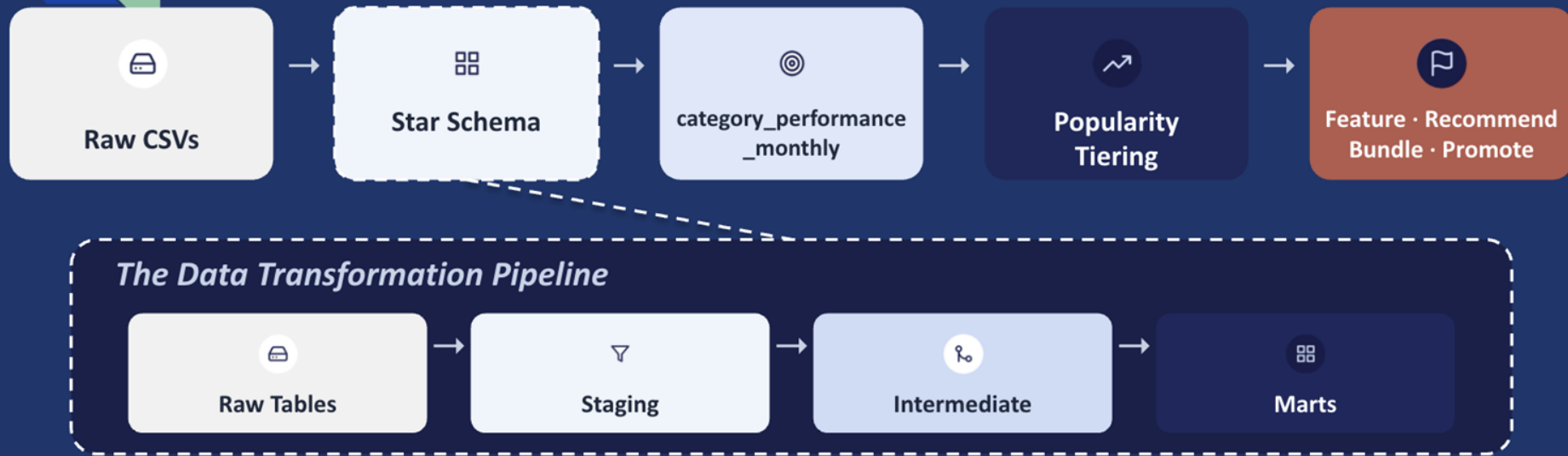


Under 1% of the free allowance used

Total BigQuery bill: \$0.00 / month — spend only starts past 10 GB stored or 1 TB scanned, roughly 50x our current footprint



From Pipeline to Insightful Decision



Each layer only ever depends on the layer before it. There's a single, traceable path from raw data to the final tables.

Data Transformation Pipeline



Single source of truth

One mart feeds every number in the executive deck.



Testable at every layer

Each transformation stands on its own and can be checked.



Traceable end to end

Every figure traces back to one raw file.

Accuracy first, because product recommendations ride on it

We prioritized data accuracy because these transformations directly underpin the critical KPIs, such as customers engagement and revenues.

Scalability

Built to keep pace with the demands of a dynamic business environment.

- Supports a fast-changing catalogue, allowing new transformation requirements to be added without reworking the existing pipeline.
- Accommodate future requirements: e.g. The Order Review capability has been deferred and can be introduced later, once sentiment analysis of review text and comments is implemented.

Staging - Cleaning

- Creates one staging table per raw file. Only renaming and type-casting. No joining of tables and application of business logic.
- **CAST** for price & freight value so any failed conversion throws an error and stops the query.
- **SAFE_CAST** for dates, where a missing delivery date simply becomes null instead of breaking the load.
- A blank category becomes a true NULL but not an empty
- string to distinguish genuinely blank from source.

Row	category_name_raw
1	null
2	bebes
3	null
4	climatizacao
5	automotivo
6	automotivo

BigQuery check: blank categories land as true NULL, never an empty string.

Intermediate - Shared logic for single computation

Denormalisation occurs in this layer are working with 6 tables, where tables are combined into flatter, easier-to-query shapes. Logic needed by two or more marts is built once here and shared.

int_order_items_enriched



One row per order item record

`int_order_items_enriched` takes the raw list of order line items and attaches an English category name to each one, by **left-joining** through the products table and the translation table.

It exists so that step only has to be done once, in one place, rather than every downstream model re-doing that same lookup itself.

int_order_payments_agg



One row per order

`int_order_payments_agg` combines an order's payments, which can span several rows (installments, vouchers, multiple cards) into totals: amount paid, number of methods used, and the main payment method because we don't require the payment pattern at a granular level.

order_items

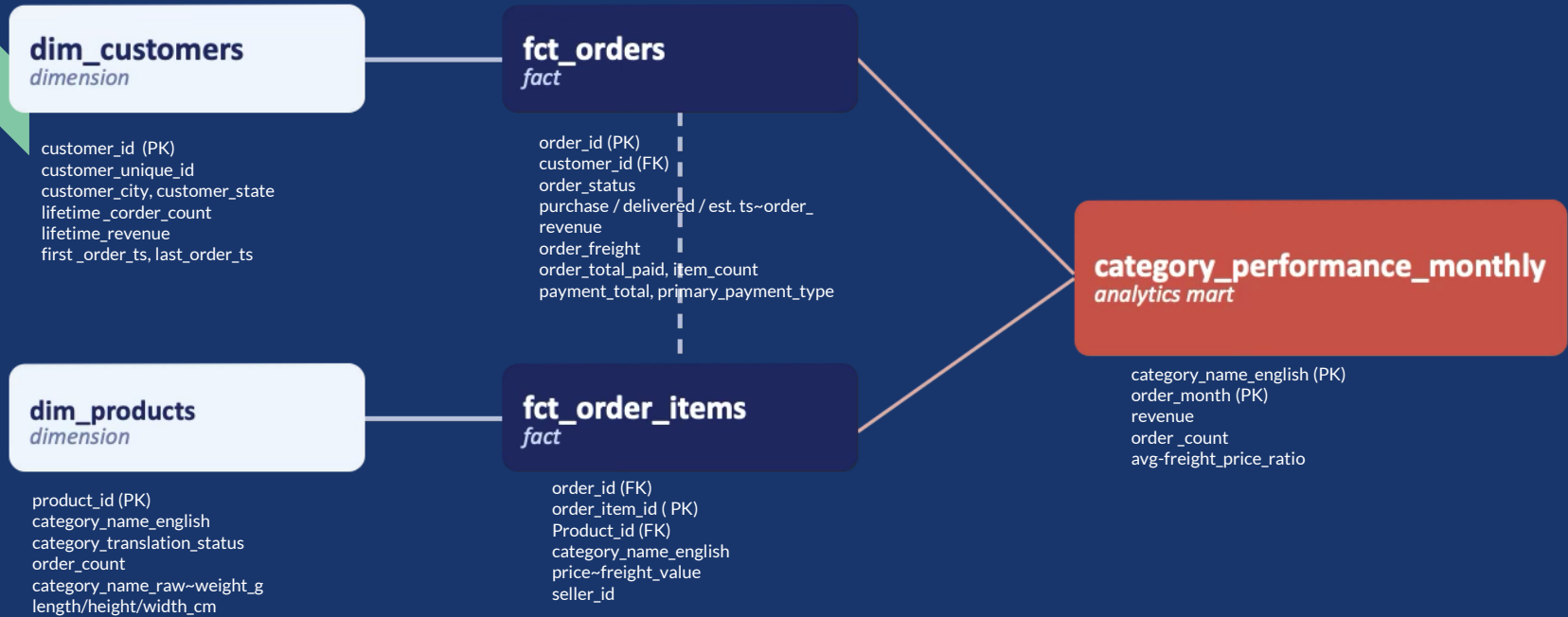
order_id	product_id	price
ord_1001	prod_A	49.90
ord_1002	prod_B	129.00
ord_1003	prod_A	15.50



products

product_id	product_category_name
prod_A	beleza_saude
prod_C	eletronicos

Mart - Business-Ready Star Schema



- dim_customers / dim_products contains secondary descriptive attributes.
- fct_orders / fct_order_items measure the business event, at two grains — order and order line.
- category_performance_monthly aggregates both fact tables into the one table that surfaces popular products and feeds the consumer-matching analysis.



Choice of tool GX and DBT

GX

Runs via Python (Pandas/Spark) or SQL backends

Can configuration table , validation through code

Modularity, modify and rerun test validation

Dedicated, enterprise-grade data quality framework

Data Testing with Great Expectations

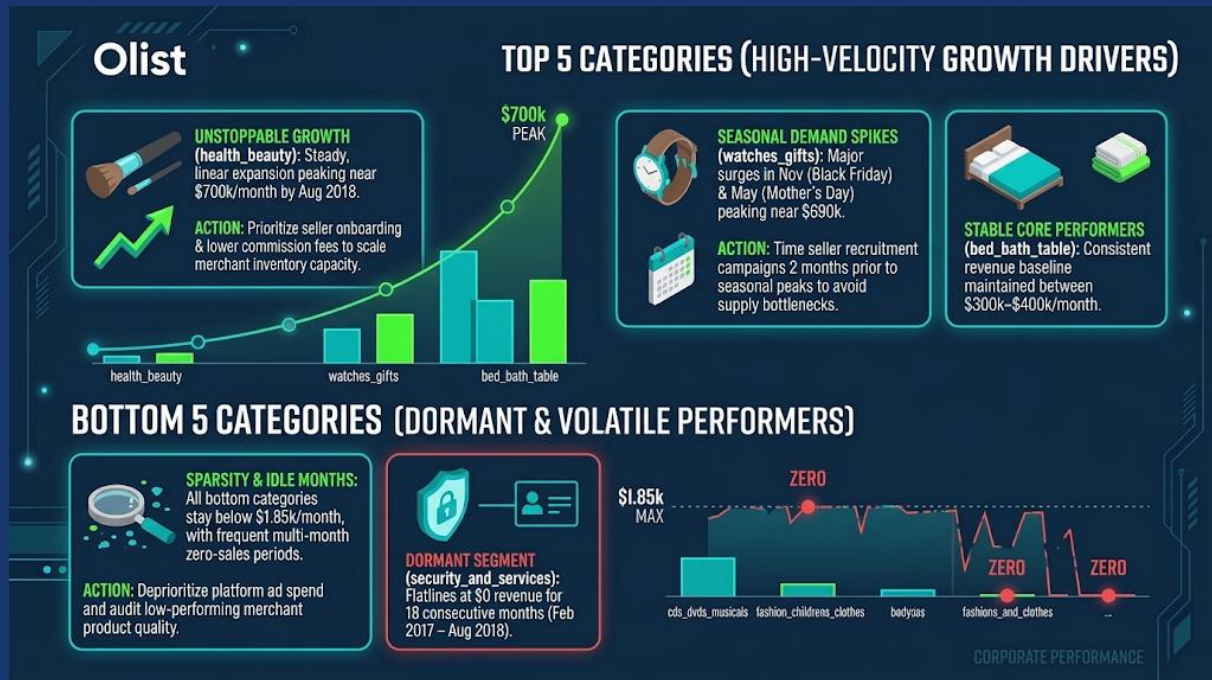
DATA TESTING WITH GREAT EXPECTATIONS

8-Step Quality Workflow

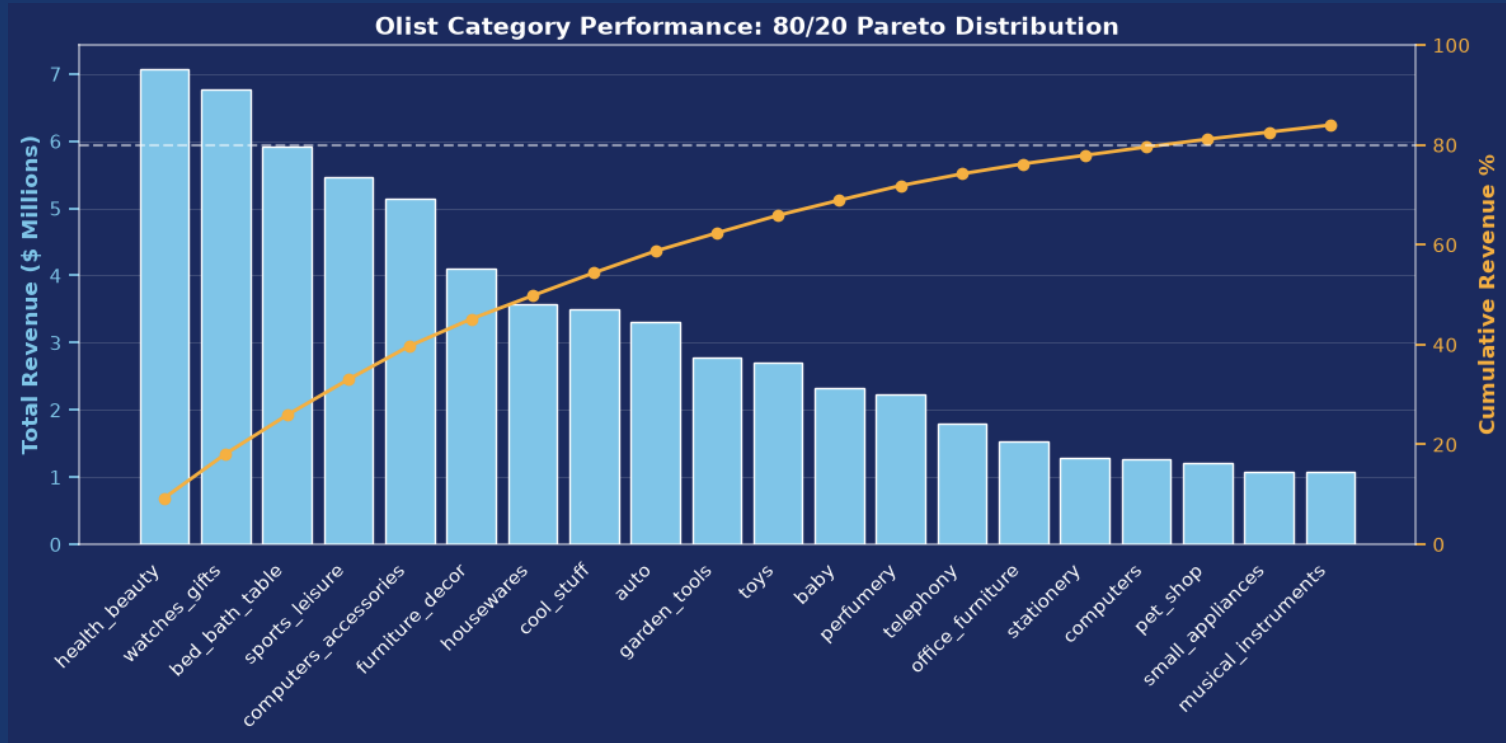


Case 1: Inventory Revenue Analysis

The 80/20 Pareto Distribution: Focus on the 20 Categories of the inventory that generates 80% of the revenue.

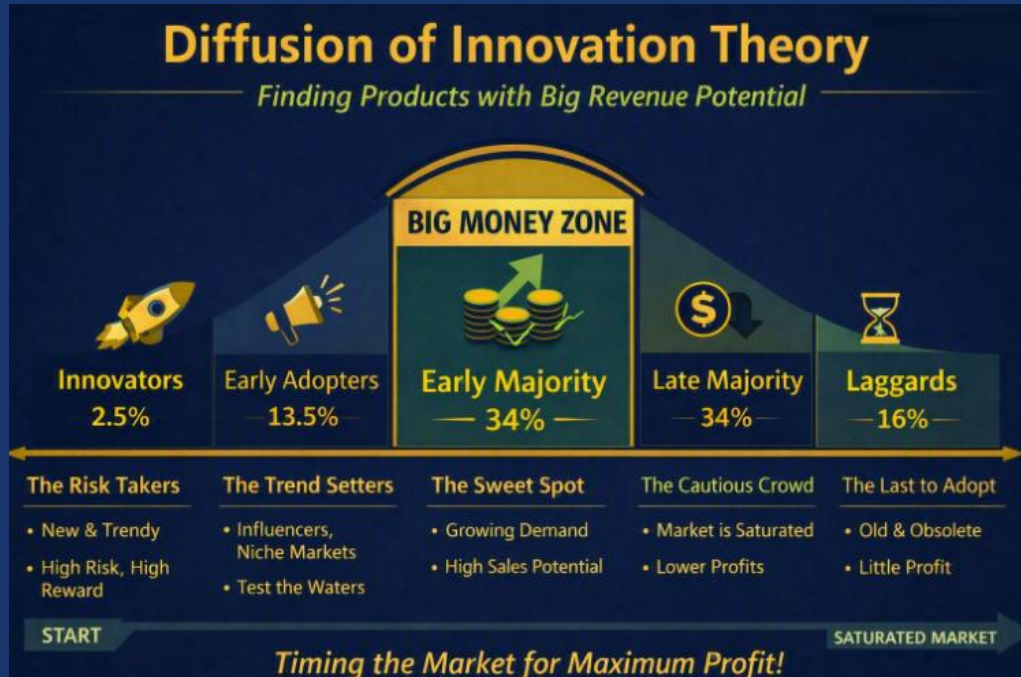


Application - Matplotlib visualisation



Case 2: Diffusion of Innovation

Approach: To observe the lifecycle trends of products over a period of time and determine its future trajectory.



AREAS OF INTEREST

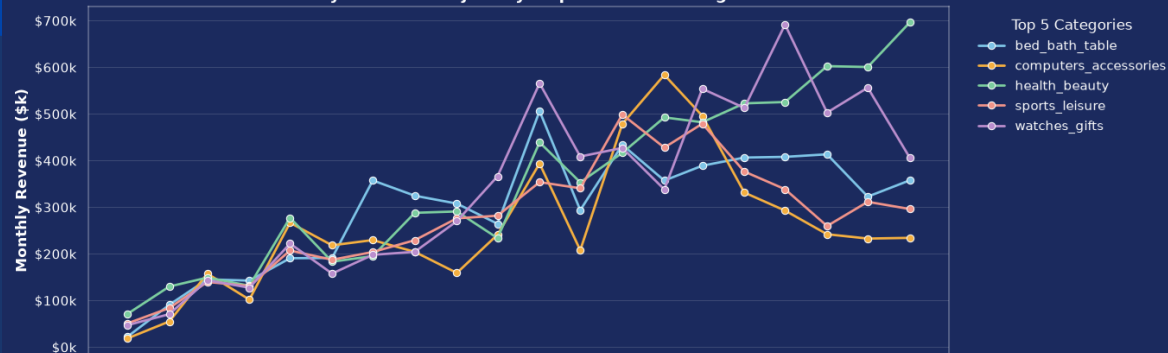
01. Innovators
02. Early Adopters
03. Big Money Zone

AREAS OF CAUTION

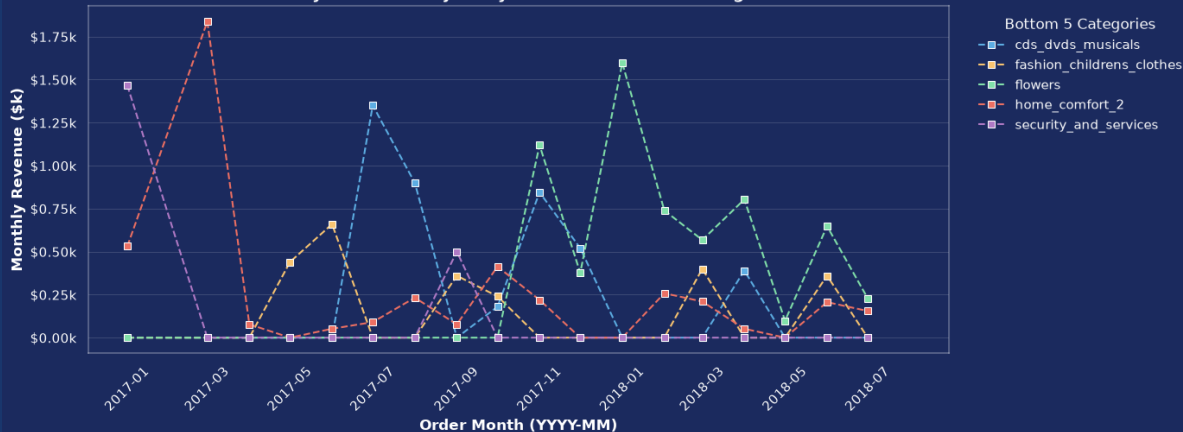
01. Late Majority
02. Laggards

Application - Monthly Category Revenue Trends

Monthly Revenue Trajectory: Top 5 Product Categories



Monthly Revenue Trajectory: Bottom 5 Product Categories



MONTHLY REVENUE TRAJECTORY

STRATEGIC OUTLINE

01. Identify innovators and early adopters for future potential revenue.
02. Worth to invest a more significant capital for future returns.
03. Identify late majority and laggard categories.
04. Clear existing inventory while limiting future inventory.



How it Addresses our Business Case

- Identifies and recommends popular products to consumers, along with trending items
- Match consumers with items they have a greater propensity to purchase
- Drives sustained platform usage and ensure the products are constantly up to date with consumer demand
- Creation of a “Bargain basement” for products in their later stage of their lifecycle to entice consumers to clear out the inventory on these categories



Thank you!